Hardware

Chassis: Traxxas68086 Slash 4x4 Platinum Ultimate

Main computer: NVIDIA Jetson Orin Nano

LiDAR: Hokuyo 10LX

Motor: Traxxas Velineon

VESC: VESC 6MK VI

IMU (built in VESC)

Servo: Digital Servo

Battery: LiPo 3S*1

Software

The software stack is a modular, ROS-based autonomous racing architecture that supports two independent driving modes: a modular pipeline following the **Perception-Planning-Control structure**, and an **end-to-end mode based on TinyLidarNet**. Only one mode runs at a time. The modular pipeline comprises state estimation, opponent estimation, planning, and control modules, described below, while the end-to-end mode is described at the end of this section.

For state estimation, the wheel odometry from the VESC is fused with IMU measurements to estimate the vehicle's velocity and yaw rate. The vehicle pose is estimated with respect to a pre-mapped racetrack, and the final vehicle state is provided in both Cartesian and Frenet coordinate frames. This state is used by all downstream modules.

For opponent estimation, the 2D LiDAR scan is segmented into obstacle clusters using the Adaptive Breakpoint method. Opponent-vehicle candidates are detected through track filtering, size filtering, and rectangle fitting. The detected objects are then passed to the tracking module, which distinguishes static obstacles from dynamic opponent vehicles and estimates the opponent vehicle's position and velocity.

For planning, the vehicle follows a precomputed global racing line during nominal driving. When an opponent is detected, the local planner generates an overtaking trajectory using a spline-based method: it determines the free space to the left and right of the opponent, places an overtaking apex together with pre-apex and post-apex points, and fits a cubic spline through them. The resulting trajectory is checked against track boundary and safety margin constraints before it is used as the final avoidance path. A planning state machine selects the driving behavior according to the current race situation: the vehicle can follow the global racing line, trail behind the opponent, or execute an overtaking maneuver. If no valid overtaking trajectory is available, it remains in trailing mode and keeps a safe distance behind the opponent.

The control module computes the steering and speed commands that track the trajectory supplied by the planner, using the estimated vehicle state and the distance to the opponent. Lateral control is performed by a MAP controller, with a Pure Pursuit controller available as an alternative configuration. A Follow-The-Gap controller is also implemented as a reactive fallback: it steers the vehicle toward the largest collision-free gap directly from the 2D LiDAR scan, and is used when state estimation becomes unreliable and the vehicle can no longer track the trajectory safely.

As an alternative to the modular pipeline, the stack also implements an end-to-end driving mode based on **TinyLidarNet**, a lightweight 1D convolutional neural network. In this mode, the raw 2D LiDAR scan is fed directly to the network, which predicts the steering and speed commands without explicit state estimation, mapping, or planning.